

PENN YAN, NY, USA

WMO: 725194

Lat: 42.643N

Lon: 77.056W

Elev: 903

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54778

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	6.9	-7.0	3.9	4.0	-2.1	5.1	9.1	26.1	28.9	23.8	28.5	7.9	210	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.9	72.1	85.9	70.2	82.9	69.1	74.4	84.1	72.9	82.1	71.4	79.7	8.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	121.3	79.4	70.1	114.6	77.5	68.6	108.7	76.3	38.4	83.7	37.0	82.4	35.7	79.6	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.6	16.7	-2.4	93.4	4.4	2.7	-5.6	95.3	-8.2	96.9	-10.7	98.4	-13.9	100.4	
			DB												
			WB	-3.3	77.0	4.3	1.7	-6.4	78.2	-8.9	79.2	-11.3	80.1	-14.3	81.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	25.4	26.7	35.1	46.5	58.6	67.1	71.5	69.9	63.6	52.1	41.2	31.4
	DBStd	18.52	11.36	10.30	10.88	9.40	8.58	6.72	5.53	5.40	7.42	8.74	8.92	9.10
	HDD50	3041	765	654	478	175	19	0	0	0	2	81	288	579
	HDD65	6382	1229	1073	928	558	237	55	9	17	113	408	714	1041
	CDD50	2753	2	2	15	70	285	512	667	618	409	147	24	2
	CDD65	620	0	0	0	4	39	117	211	169	71	9	0	0
	CDH74	4706	0	0	2	74	382	922	1644	1139	491	52	0	0
	CDH80	1357	0	0	0	17	102	274	513	308	137	6	0	0
	WSAvg	7.2	8.8	8.6	8.3	8.0	6.6	6.1	5.8	5.3	5.7	6.9	8.2	8.6
	PrecAvg	34.6	1.8	1.8	2.3	2.8	3.1	3.3	3.4	3.4	3.5	3.0	2.6	2.4
Precipitation	PrecMax	42.1	3.9	3.5	6.1	5.9	6.1	5.7	9.1	6.0	5.7	6.8	4.2	4.3
	PrecMin	28.2	0.7	0.8	0.6	1.0	0.8	0.9	0.9	1.4	1.5	0.9	0.8	1.0
	PrecStd	3.6	0.7	0.8	1.2	1.3	1.5	1.4	1.7	1.3	1.3	1.5	0.9	0.9
	PrecStd	3.6	0.7	0.8	1.2	1.3	1.5	1.4	1.7	1.3	1.3	1.5	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	60.7	70.8	82.1	87.4	90.7	91.4	91.0	89.3	79.9	67.4	61.6
		MCWB	52.1	50.8	57.4	63.0	68.2	72.6	73.6	73.0	71.0	66.1	55.9	54.9
	2%	DB	51.8	51.8	62.8	74.2	82.9	87.0	88.7	86.9	83.9	74.3	63.2	53.0
		MCWB	46.8	44.5	51.8	58.3	65.9	70.3	72.7	71.9	69.8	62.7	54.6	47.4
	5%	DB	45.5	45.9	56.8	67.8	78.6	82.8	85.5	83.0	79.8	69.9	59.4	48.4
		MCWB	41.4	40.0	49.2	54.3	64.2	68.9	70.9	70.0	67.7	61.7	51.5	44.1
	10%	DB	40.6	40.8	51.8	63.1	73.4	79.4	82.3	80.7	75.4	65.2	55.1	44.4
		MCWB	36.5	36.0	44.7	52.6	61.2	66.8	69.4	68.7	65.8	58.1	48.8	40.2
	0.4%	WB	53.4	53.3	58.8	65.5	71.4	75.2	77.1	76.3	74.3	69.5	59.2	54.8
		MCDB	57.2	59.2	67.7	78.6	83.0	85.6	88.0	84.9	84.0	75.4	64.1	60.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.5	45.3	54.5	60.0	68.4	72.8	74.6	73.9	72.0	65.7	56.1	48.9
		MCDB	50.1	51.3	61.7	70.7	78.9	83.2	84.9	82.4	80.6	72.1	61.8	52.4
	5%	WB	41.5	40.8	49.7	56.9	66.1	70.9	72.9	72.3	69.7	62.2	53.0	44.4
		MCDB	45.2	45.6	55.9	66.3	75.1	79.9	82.1	80.3	76.7	68.1	58.3	48.3
	10%	WB	37.0	36.0	45.2	53.2	63.7	68.9	71.3	70.6	67.6	59.1	49.7	40.5
		MCDB	40.4	40.5	51.1	61.6	71.4	76.8	79.4	77.6	73.4	64.8	54.9	44.2
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.6	16.6	19.5	20.1	18.2	18.2	17.9	18.6	15.8	14.3	11.6
		MCDBR	18.0	20.8	24.5	29.1	25.2	22.1	20.8	20.6	21.2	21.0	20.7	16.7
		MCWBR	15.4	16.3	16.0	16.3	12.6	10.0	8.9	8.7	9.7	12.3	14.1	13.8
	5% WB	MCDBR	17.4	20.4	22.7	26.1	22.2	19.6	18.0	17.5	18.4	18.8	18.5	16.2
		MCWBR	15.7	17.1	16.0	16.5	12.2	10.3	8.9	8.8	9.7	12.6	14.5	14.2
Clear-Sky Solar Irradiance	taub		0.301	0.317	0.341	0.371	0.390	0.438	0.440	0.429	0.370	0.352	0.310	0.291
	taud		2.370	2.327	2.363	2.336	2.322	2.208	2.214	2.231	2.385	2.433	2.511	2.467
	Ebn at Noon		265	279	286	284	280	266	264	263	273	264	262	261
	Edh at Noon		25	31	34	38	40	45	44	42	33	28	22	21
	RadAvg		429	696	1048	1398	1685	1787	1820	1628	1303	803	510	344
All-Sky Solar Radiation	RadStd		38	88	100	156	167	159	144	94	126	83	67	51

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A

Nomenclature: See separate page